

Beyond the Static Gradient: Electrons on Helium as a Time-Dependent-Field Sensor for Particle Physics

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Reference [?] proposed a pair of Coulomb-entangled electrons on superfluid helium, prepared in the motional/spin singlet, as a differential sensor for particle-physics signals, demonstrating a factor-of-two quantum-Fisher-information gain over an independent-spin Ramsey benchmark for a *static* magnetic-field gradient. That work explicitly deferred time-dependent fields to future study. This project takes up that extension: we generalize the singlet-triplet sensor Hamiltonian to an oscillating or transient coupling $\lambda(t)$ and ask which particle-physics targets are best matched to it. A literature survey of running and proposed experiments — oscillating axion/axion-like-particle (ALP) dark-matter backgrounds (CASPER, ABRACADABRA, DM Radio), dark-photon dark matter, network-based domain-wall/transient searches (GNOME), and precision electric-dipole-moment (EDM) measurements — is used to identify which regime (resonant narrowband, broadband stochastic, or impulsive) the eHe platform’s characteristic frequency scale and common-mode-rejection architecture could plausibly compete in. This is a proposal: we outline the extended formalism, the candidate target channels with their existing experimental sensitivity reach, and the open questions that must be resolved before a concrete protocol can be simulated.

I. INTRODUCTION

Two Coulomb-entangled electrons on superfluid helium, prepared in the motional/spin singlet, were shown in Ref. [?] to act as a differential quantum sensor: blind to common-mode fields and maximally responsive to a *difference* between the two trap sites, quantified through the quantum and classical Fisher information and benchmarked against an independent-spin Ramsey scheme. The entangled states and the numerical machinery used to obtain them (Hartree/CI on a sine-DVR basis, configurations I/II/III) were established in Ref. [?], with time-dependent *control* of the same states realized as high-fidelity two-qubit gates in Ref. [?]; Ref. [?] also grounds the sensing readout in the real resonator/charge-readout hardware demonstrated in Refs. [? ?].

Reference [?] is explicit about its scope: “we limit ourselves in this work to the detection of a static magnetic-field gradient across the two wells,” and states that “we will address time-dependent fields in future studies.” The same paper’s discussion section already gestures at what those time-dependent targets might be — “the extremely weak, slowly oscillating fields associated with ultralight dark-matter candidates such as axions or axion-like particles” [? ? ?], entangled sensor networks for dark-matter and axion signals [? ?], ultrafast transients from milli-charged particles or hidden-sector bosons, and rare-event searches including electric dipole moments — without developing any of them quantitatively. The paper’s own sensor-network outlook (Sec. 2.9) further distinguishes two coupling channels that a single eHe platform could in principle host: the singlet, optimal for *gradient* signals (generator G_-), and Bell/GHZ-type states, optimal for *common-mode* oscillating signals (generator G_+) such as “a weak oscillating field from an axion-like background.”

This project asks two questions left open by Ref. [?]: (i) how does the static-field QFI/CRB formalism of Sec. IV of that paper generalize to a periodic or transient $\lambda(t)$, and (ii) among the particle-physics targets it names only qualitatively, which one is quantitatively the best match for the eHe platform’s trap geometry, coherence budget, and $G_{\pm}$ readout modes, given where existing dedicated experiments already sit in sensitivity and frequency/mass coverage. We do not yet have a settled answer to either question; the goal of this manuscript, at this stage, is to lay out the extended model and a literature-grounded comparison rather than to report a result.

II. MODEL

A. From a static to a time-dependent coupling

The starting point is the effective two-level (singlet-like/triplet-like) model of Ref. [?],

$$H_{\text{eff}}(t) = E_g |g\rangle\langle g| + E_e |e\rangle\langle e| + \lambda(t) (|g\rangle\langle e| + |e\rangle\langle g|), \quad (1)$$

where the original proposal fixes $\lambda = (g\mu_B \delta B/2) M_{ge}$ for a *static* field gradient $\delta B = B_L - B_R$, with matrix element $M_{ge} = \langle e | S_L^z - S_R^z | g \rangle$. The natural generalization replaces the static δB with a time-dependent source term. Two limiting forms recur throughout the axion/dark-photon haloscope literature and map directly onto candidate targets identified below:

- **Narrowband/resonant:** $\delta B(t) = \delta B_0 \cos(\omega_a t + \phi)$, appropriate for a coherent ultralight bosonic dark-matter field of mass $m_a = \hbar\omega_a$ coherent over the local dark-matter coherence time $\tau_c \sim 10^6/\omega_a$ — the regime targeted by CASPER [?], ABRACADABRA [? ?], and DM Radio [?].

- **Impulsive/transient:** $\delta B(t) = \delta B_0 f(t/\tau)$ for a short pulse of duration τ crossing the trap, appropriate for a milli-charged particle, hidden-sector boson, or domain-wall crossing — the regime targeted by network searches such as GNOME [? ?].

In both cases the quantum and classical Fisher information of Sec. IV of Ref. [?] must be re-derived for a coupling with explicit time dependence, replacing the single scalar parameter δB with either an amplitude-and-phase pair $(\delta B_0, \phi)$ at fixed known ω_a (matched-filtering regime, as in resonant haloscopes) or a fully unknown waveform (change-point/transient-detection regime, as in network searches). Neither extension is a straightforward relabeling of the existing static-parameter QFI/CRB result; both require revisiting which quantity is actually being estimated.

B. Common-mode vs. gradient coupling

Whether a given target couples through the gradient generator G_- (as in the original static proposal) or the common-mode generator G_+ depends on the physical origin of the signal. An oscillating axion-induced pseudomagnetic field sourced by the galactic dark-matter background is, to leading order, spatially uniform on trap length scales ($\sim \mu\text{m}$), favoring a G_+ (Bell/GHZ) readout as anticipated in Ref. [?]'s outlook; a signal from a particle traversing (or passing near) one of the two wells, or a topological-defect/domain-wall crossing with a finite spatial extent, instead favors the G_- (singlet, gradient) readout already developed for the static case. Dark-photon dark matter can source either an effectively common-mode oscillating field (through kinetic mixing with the photon, felt uniformly by both wells) or, if the dark-photon Compton wavelength becomes comparable to the trap baseline, a genuine gradient signal — a distinction not present in the original static proposal and requiring its own treatment.

III. CANDIDATE TARGET CHANNELS (TO BE EVALUATED)

- **Oscillating axion/ALP dark-matter background.** The best-motivated extension of Ref. [?]'s own framing: axion/ALP dark matter couples to electron (and nucleon) spins as an effective, slowly oscillating Zeeman-like term [? ?]. Dedicated experiments (CASPER-Electric/Gradient [?], the inductively coupled ABRACADABRA-10 cm [? ?], and the lumped-element DM Radio pathfinder [?]) already cover a broad low-mass window ($\sim 10^{-14}$ – 10^{-6} eV); recent work also shows that the same NMR/comagnetometer-style apparatus is simultaneously sensitive to the axion-photon coupling and to kinetically mixed dark photons [?]. The

open question is whether the eHe platform's trap-baseline and coherence-time scale places it in a genuinely uncovered mass window, or whether it would only reproduce existing haloscope sensitivity with a different (entanglement-enhanced) prefactor.

- **Dark-photon dark matter.** A kinetically mixed dark photon sources an oscillating effective electric or magnetic field inside a shielded volume, detectable with the same spin-precession or charge-coupled apparatus used for axions, with a distinguishable frequency/coupling signature [?]. Since the eHe platform already has both a spin (G_-/G_+ Zeeman) and a charge ($n_L - n_R$, cf. the **majorana-two-electron-sensor** project's charge channel) coupling route, this may be the channel best suited to exploiting the platform's dual readout.
- **Network transient/domain-wall searches.** GNOME-style searches [? ?] correlate transient signals across a global network of comagnetometers to detect topological-defect dark matter crossing the Earth. An array of entangled eHe sensors (cf. Ref. [?] Sec. 2.9 and the **rl-agentic-quantum-control** project's optimization machinery) could in principle contribute a single, high-coherence node to such a network, or motivate an analogous laboratory-scale multi-trap correlation search; the impulsive-signal QFI/CRB treatment above is the theoretical prerequisite.
- **Electric dipole moment (EDM) precision channel.** A static electric field applied across the trap, combined with a comagnetometer-style oscillating readout, is the standard EDM protocol [?]; entangled-sensor EDM proposals in trapped-atom systems report projected sensitivity below 10^{-30} e·cm. This channel is intermediate between the original static-gradient proposal (a static source field) and the oscillating targets above (an oscillating readout is still required to reject systematics), and would need its own assessment of whether the eHe platform's achievable E -field and coherence time are competitive.

- **Broader spin-dependent short-range forces.** Reviewed in Ref. [?], “fifth-force” searches for exotic spin-dependent interactions are a natural extension of the gradient (G_-) channel already built for Ref. [?], since they are inherently short-range, differential signals rather than galactic-background common-mode ones — potentially the lowest-effort extension of the existing static formalism, trading a particle-physics dark-matter motivation for a tabletop BSM-force one.

Each channel implies a different generator (G_- vs. G_+), a different estimation-theoretic regime (resonant

narrowband vs. transient), and a different existing-experiment baseline to beat or complement; narrowing this list — using, at minimum, an order-of-magnitude comparison between the eHe platform’s coherence-time-limited sensitivity and the published exclusion curves of Refs. [? ? ? ? ? ? ?] — is the first concrete task for this project. A broader Snowmass-style survey of quantum sensors for high-energy physics [?] is used as a cross-check that no obviously more appropriate existing platform is being overlooked.

IV. OPEN THEORETICAL QUESTIONS

- How does the quantum Cramér–Rao bound of Ref. [?] generalize when the parameter of interest is an oscillation amplitude at a known (or partially known) frequency, versus an unknown transient waveform? Is a Bayesian/matched-filter treatment, as used in the haloscope literature, more natural here than the single-shot static-parameter QFI used previously?
- Does the factor-of-two entanglement advantage found for the static gradient survive, strengthen, or degrade once averaged over a stochastic (finite dark-matter-coherence-time) oscillating source, as opposed to a fixed classical field?
- What eHe trap coherence time and baseline (electron–electron separation) would be needed to reach a mass/frequency window not already covered by CASPER, ABRACADABRA, DM Radio, or GNOME, and is that window physically achievable given the device parameters used in Ref. [? ?]?
- For the network/transient channel, does the existing single-pair sensor need to be embedded in a multi-trap array (per Ref. [?] Sec. 2.9) before a

meaningful comparison to GNOME- style searches is even possible, or is a single high-coherence node already a useful complementary data point?

V. RESULTS

Not yet available — this section will report the channel-comparison order-of-magnitude estimates and, for the strongest candidate, a time-dependent QFI/CRB derivation analogous to Sec. IV of Ref. [?].

VI. DISCUSSION AND OUTLOOK

If a viable channel is identified, next steps are: (i) extend the CI/Hartree machinery of Ref. [?] and the numerical toolkit described in the project’s `ci-dvr-hartree-numerics` skill to propagate the entangled state under an oscillating or transient $\lambda(t)$; (ii) derive the corresponding time-dependent quantum and classical Fisher information, including the effect of finite dark-matter coherence time where relevant; (iii) compare the achievable sensitivity, as a function of trap baseline and coherence time, against the published exclusion curves for the channel chosen; (iv) assess whether RL-based trap optimization (see the companion `rl-agentic-quantum-control` project) can be re-targeted from the static-gradient reward used in Ref. [?] to an oscillating-signal reward. This project is complementary to, not a replacement for, the `majorana-two-electron-sensor` project, which explores a condensed-matter (rather than particle-physics) target for the same differential-sensor machinery.

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Placeholder.